UNIVERSITY OF INFORMATION TECHNOLOGY
FACULTY OF INFORMATION SYSTEMS

Chapter 1

INTRODUCTION

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Introduction

- Traditional database applications
  - Store textual or numeric information
- Multimedia databases
  - Store images, audio clips, and video streams digitally
- Geographic information systems (GIS)
  - Store and analyze maps, weather data, and satellite images
- Data warehouses and online analytical processing (OLAP) systems
  - Extract and analyze useful business information from very large databases
  - Support decision making
- Real-time and active database technology
  - Control industrial and manufacturing processes

Thời khoá biểu

THỜI KHOÁ BIỂU HỌC KỲ 2 NĂM 2020 - 2021

MãMH	Lớp	Tên MH	KhoaQL	SốTC	Thứ / Tiết / Phòng	GV	SS	ĐãĐK
ACCT5123	ACCT5123.L21.CTTT	Hoạch định nguồn lực doanh nghiệp	HTTT	3	Thứ 2, Tiết 123, P.C214	Đỗ Phúc	50	44
ACCT5123	ACCT5123.L21.CTTT.1	Hoạch định nguồn lực doanh nghiệp (HT1 - Học cách 2 tuần)	HTTT	0	Thứ 3, Tiết 67890, P.B4.04	Huỳnh Đức Huy	25	22
ACCT5123	ACCT5123.L21.CTTT.2	Hoạch định nguồn lực doanh nghiệp (HT1 - Học cách 2 tuần)	HTTT	0	Thứ 3, Tiết 67890, P.B4.04	Huỳnh Đức Huy	25	22
CS2133	CS2133.L21.CTTT	Khoa học máy tính II	HTTT	3	Thứ 6, Tiết 678, P.E44	Ngô Đức Thành	40	39
~~CS2133~~	~~CS2133.L21.CTTT.1~~	~~Khoa học máy tính II (HT1 - Học cách 2 tuần)~~	~~HTTT~~	~~1~~	~~Thứ 7, Tiết 12345, P.B3.02~~	~~Đỗ Văn Tiến~~	~~20~~	~~20~~
CS2133	CS2133.L21.CTTT.2	Khoa học máy tính II (HT1 - Học cách 2 tuần)	HTTT	1	Thứ 7, Tiết 12345, P.B3.02	Đỗ Văn Tiến	20	19
CS2133	CS2133.L22.CTTT	Khoa học máy tính II	HTTT	3	Thứ 5, Tiết 678, P.E44	Lê Minh Hưng	40	39
~~CS2133~~	~~CS2133.L22.CTTT.1~~	~~Khoa học máy tính II (HT1 - Học cách 2 tuần)~~	~~HTTT~~	~~1~~	~~Thứ 5, Tiết 12345, P.C202~~	~~Đỗ Văn Tiến~~	~~20~~	~~20~~
CS2133	CS2133.L22.CTTT.2	Khoa học máy tính II (HT1 - Học cách 2 tuần)	HTTT	1	Thứ 5, Tiết 12345, P.C202	Đỗ Văn Tiến	20	19

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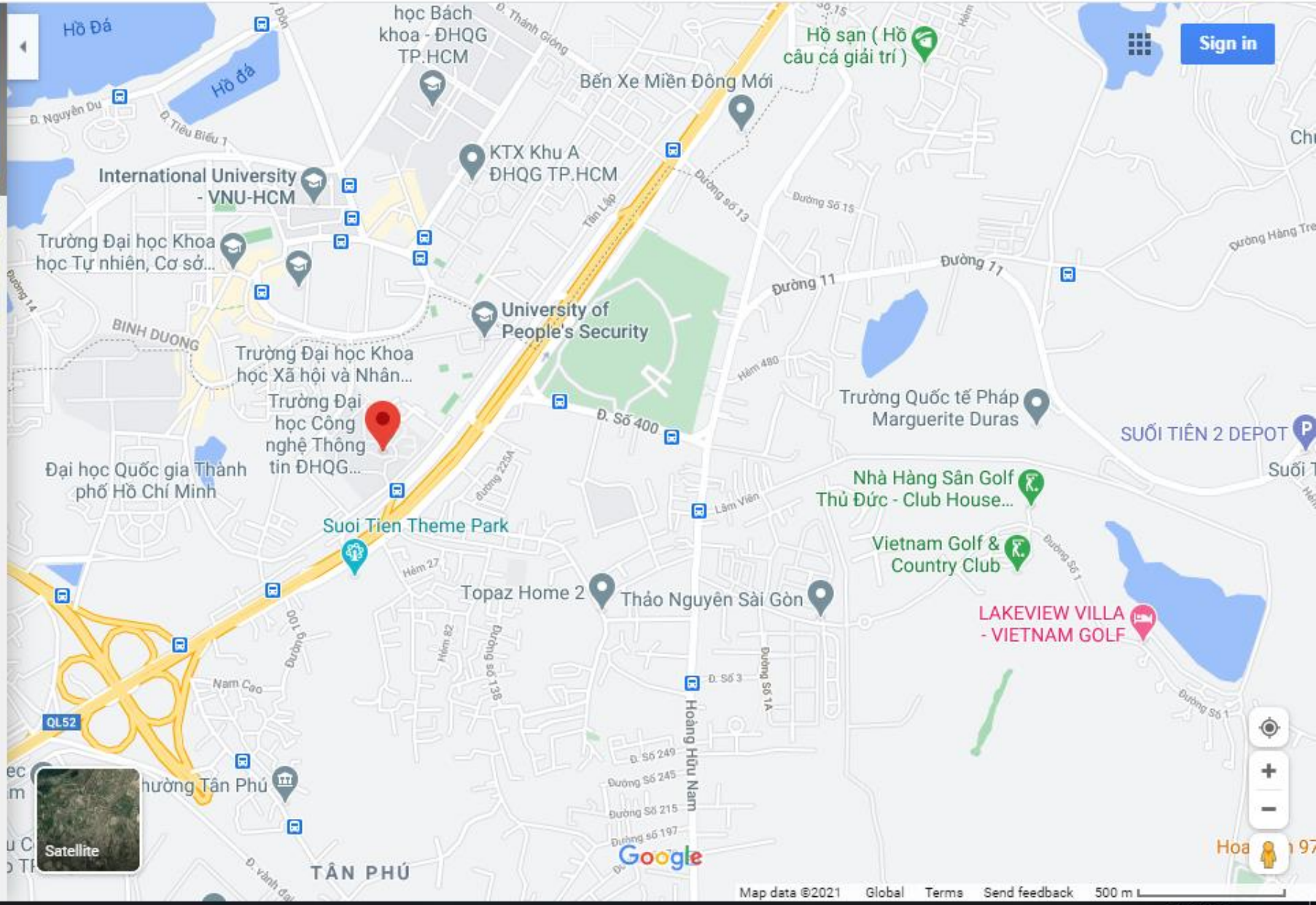

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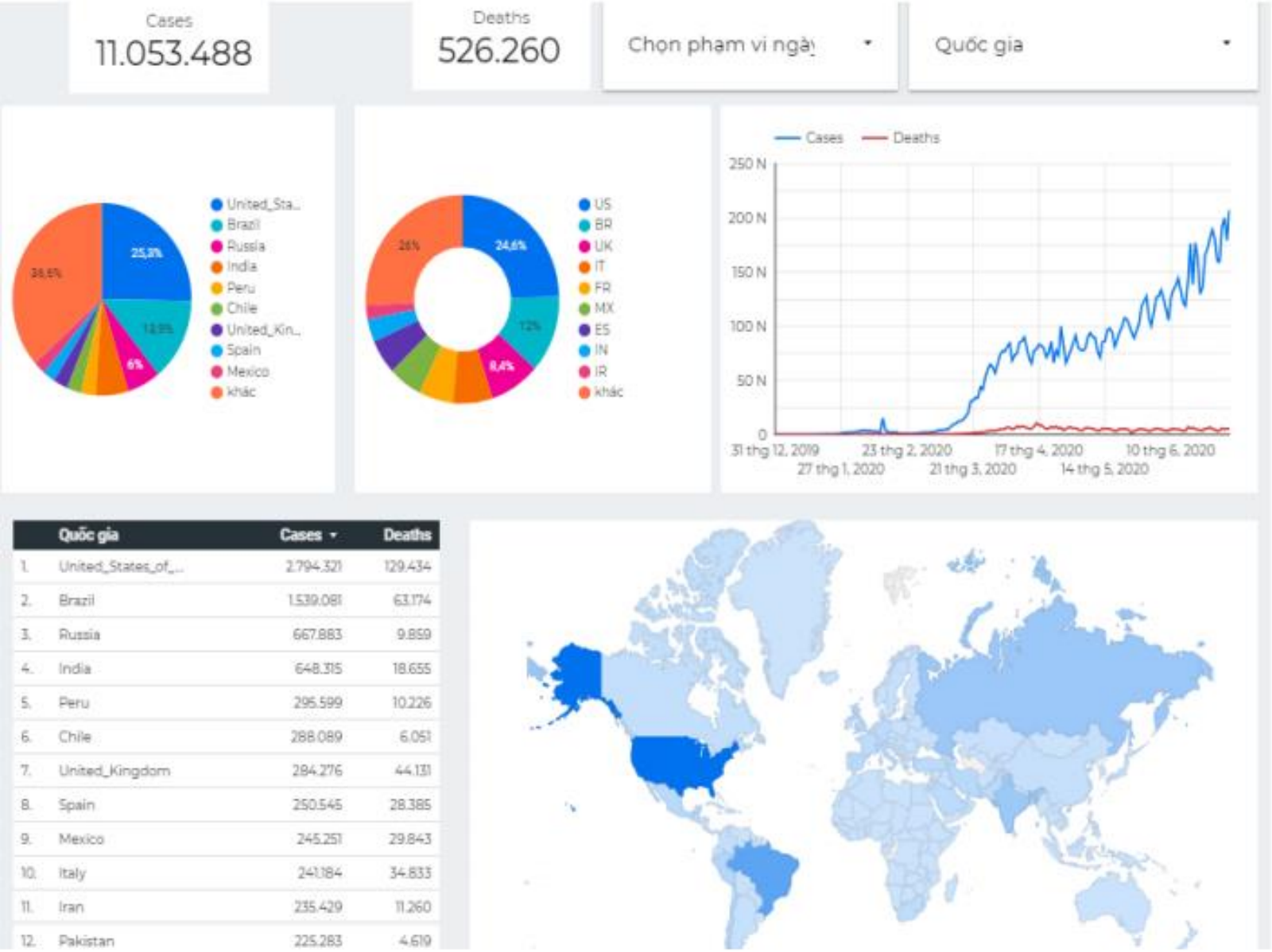

	Quốc gia	Cases	Deaths
1.	United_States_of_...	2.794.321	129.434
2.	Brazil	1.539.081	63.174
3.	Russia	667.883	9.859
4.	India	648.315	18.655
5.	Peru	295.599	10.226
6.	Chile	288.089	6.051
7.	United_Kingdom	284.276	44.131
8.	Spain	250.545	28.385
9.	Mexico	245.251	29.843
10.	Italy	241.184	34.833
11.	Iran	235.429	11.260
12.	Pakistan	225.283	4.619

Nguyen Cao Thien Tri, Chau Nhat Tien, BUILDING A TOOL FOR ANALYSING OF THE COVID-19, thesis, UIT, 2020

Introduction

- Database
  - Collection of related data
  - Known facts that can be recorded and that have implicit meaning
  - **Miniworld** or **universe of discourse (UoD)**
  - Represents some aspects of the real world
  - Logically coherent collection of data with inherent meaning
  - Built for a specific purpose

Introduction

- Example of Database
  - E-commerce systems: Tiki, Lazada,…
  - University systems: University of Information Technology, University of Science,…
  - Hospital systems: University Medical Center HCMC…
  - Healthcare systems: California Fitness & Yoga,…

Introduction

- **Database management system (DBMS)**
  - Collection of programs
  - Enables users to create and maintain a database.
  - Include processes of defining, constructing, manipulating, and sharing database among various users and applications
- **Defining** a database
  - Specify the data types, structures, and constraints of the data to be stored
- **Meta-data**
  - Database definition or descriptive information
  - Stored by the DBMS in the form of a database catalog or dictionary
- **Constructing**
  - Store the data on some storage medium

Introduction

- **Manipulating** a database
  - Query and update the database miniworld
  - Generate reports
- **Sharing** a database
  - Allow multiple users and programs to access the database simultaneously
- **Application program**
  - Accesses database by sending queries to DBMS
- **Query**
  - Causes some data to be retrieved
- **Transaction**
  - May cause some data to be read and some data to be written into the database

Introduction

- **Protection** includes:
  - System protection against hardware of software crashes
  - Security protection against unauthorized access
- **Maintain** the database system
  - Allow the system to evolve as requirements change over time

An example

- UNIVERSITY database
  - Information concerning students, courses, and grades in a university environment
- **Data records**
  - STUDENT
  - COURSE
  - SECTION
  - GRADE_REPORT
  - PREREQUISITE

An example

- Specify structure of records of each file by specifying **data type** for each **data element**
  - String of alphabetic characters
  - Number
  - Date
  - …

An example

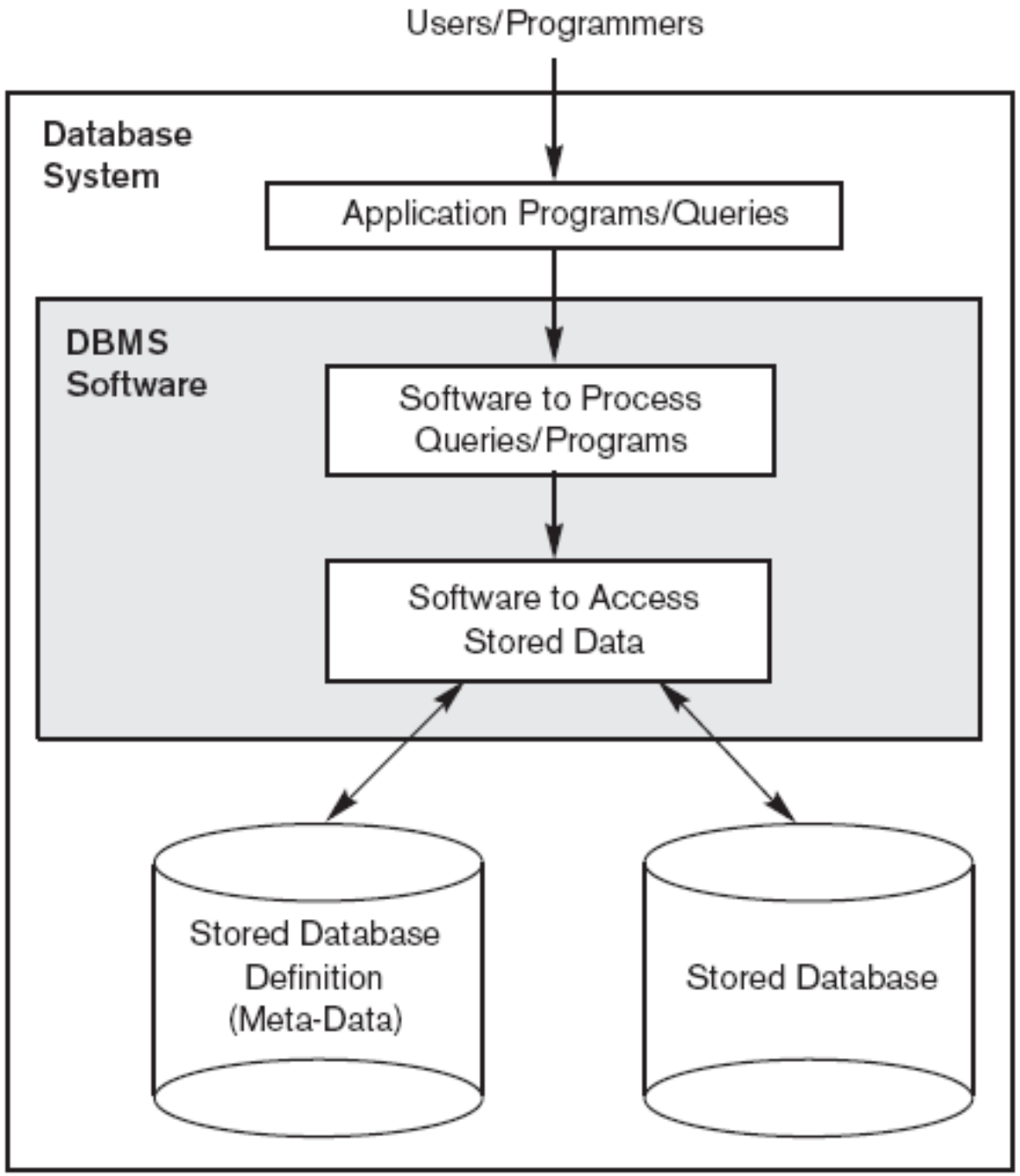

Figure 1.1
A simplified database system environment.

An example

- Construct UNIVERSITY database
  - Store data to represent each student, course, section, grade report, and prerequisite as a record in appropriate file
- Relationships among the records
- Manipulation involves querying and updating

An example

- Examples of queries:
  - Retrieve the transcript, schedule.
  - List the names of students who studied the '*Principle of Database Systems*' course offered in semester: 2, academic year: 2020-2021 and their grades
  - List the prerequisites of the '*Principle of Database Systems*' course
- Examples of updates:
  - Change the teacher of 'Smith' to 'John'
  - Create a new '*Principle of Database Systems*' course for this semester
  - Enter a grade of 'A' for 'Smith' in the *Principle of Database Systems*' course

An example

- Phases for designing a database:
  - **Requirements specification and analysis**
  - **Conceptual design (ERD – Entity Relationship Diagram)**
  - **Logical design (Relational Data model, object data model…)**
  - **Physical design**

An example

STUDENT

Name	Student_number	Class	Major
Smith	17	1	CS
Brown	8	2	CS

COURSE

Course_name	Course_number	Credit_hours	Department
Intro to Computer Science	CS1310	4	CS
Data Structures	CS3320	4	CS
Discrete Mathematics	MATH2410	3	MATH
Database	CS3380	3	CS

SECTION

Section_identifier	Course_number	Semester	Year	Instructor
85	MATH2410	Fall	07	King
92	CS1310	Fall	07	Anderson
102	CS3320	Spring	08	Knuth
112	MATH2410	Fall	08	Chang
119	CS1310	Fall	08	Anderson
135	CS3380	Fall	08	Stone

GRADE_REPORT

Student_number	Section_identifier	Grade
17	112	B
17	119	C
8	85	A
8	92	A
8	102	B
8	135	A

PREREQUISITE

Course_number	Prerequisite_number
CS3380	CS3320
CS3380	MATH2410
CS3320	CS1310

Figure 1.2
A database that stores student and course information.

Characteristics of the Database Approach

- Traditional **file processing**
  - Each user defines and implements the files needed for a specific software application → redundancy: waste storage space and get problems in data update.
- Database approach
  - Single repository maintains data that is defined once and then accessed by various users

Characteristics of the Database Approach

- Main characteristics of database approach
  - Self-describing nature of a database system
  - Insulation between programs and data, and data abstraction
  - Support of multiple views of the data
  - Sharing of data and multiuser transaction processing

Characteristics of the Database Approach

- Self-describing nature of a database system
  - Database system contains complete definition of structure and constraints
  - Meta-data
    - ✓ Describes structure of the database
  - Database catalog used by:
    - ✓ DBMS software
    - ✓ Database users who need information about database structure

Characteristics of the Database Approach

- Insulation between programs and data, and data abstraction
  - Program-data independence
    - ✓ Structure of data files is stored in DBMS catalog separately from access programs
  - Program-operation independence
    - ✓ **Operations** specified in two parts:
      - ❖ Interface includes operation name and data types of its arguments
      - ❖ Implementation can be changed without affecting the interface

Characteristics of the Database Approach

- Support of multiple views of the data
  - **View**
    - ✓ Subset of the database
    - ✓ Contains **virtual data** derived from the database files but is not explicitly stored
  - Multiuser DBMS
    - ✓ Users have a variety of distinct applications
    - ✓ Must provide facilities for defining multiple views

Characteristics of the Database Approach

- Sharing of data and multiuser transaction processing
  - Allow multiple users to access the database at the same time
  - **Concurrency control software**
    - ✓ Ensure that several users trying to update the same data do so in a controlled manner → Result of the updates is correct
  - **Online transaction processing (OLTP) application**
  - **Transaction**
    - ✓ Central to many database applications
    - ✓ Executing program or process that includes one or more database
    - ✓ **Isolation** property: Each transaction appears to execute in isolation from other transactions
    - ✓ **Atomicity** property: Either all the database operations in a transaction are executed or none are

Actors on the Scene

- **Database administrators (DBA)** are responsible for:
  - Authorizing access to the database
  - Coordinating and monitoring its use
  - Acquiring software and hardware resources
- **Database designers** are responsible for:
  - Identifying the data to be stored
  - Choosing appropriate structures to represent and store this data
- **End users**
  - People whose jobs require access to the database
- **System analysts**
  - Determine requirements of end users
- **Application programmers**
  - Implement these specifications as programs

Advantages of Using the DBMS Approach

- Controlling redundancy
  - **Data normalization**
  - **Denormalization**
    - ✓ Sometimes necessary to use **controlled redundancy** to improve the performance of queries
- Restricting unauthorized access
  - Security and authorization subsystem
  - Privileged software

Advantages of Using the DBMS Approach

- Providing backup and recovery
  - **Backup and recovery subsystem** of the DBMS is responsible for recovery
- Providing multiple user interfaces
  - **Graphical user interfaces (GUIs)**
- Representing complex relationships among data
  - May include numerous varieties of data that are interrelated in many ways

Advantages of Using the DBMS Approach

- Enforcing **integrity constraints**
  - **Referential integrity** constraint
    - ✓ Every section record must be related to a course record
  - **Key** or **uniqueness** constraint
    - ✓ Every course record must have a unique value for Course_number
  - **Business rules**
  - **Inherent rules** of the data model

Advantages of Using the DBMS Approach

- Permitting inferencing and actions using rules
  - **Deductive database systems**
    - ✓ Provide capabilities for defining deduction **rules**
    - ✓ Inferencing new information from the stored database facts
  - **Trigger**
    - ✓ Rule activated by updates to the table
  - **Stored procedures**
    - ✓ More involved procedures to enforce rules

Advantages of Using the DBMS Approach

- Additional implications of using the database approach
  - Reduced application development time
  - Flexibility
  - Availability of up-to-date information
  - Economies of scale

A Brief History of Database Applications

- Early database applications using hierarchical and network systems
  - Large numbers of records of similar structure
- Providing data abstraction and application flexibility with relational databases
  - Separates physical storage of data from its conceptual representation
  - Provides a mathematical foundation for data representation and querying

A Brief History of Database Applications

- Object-oriented applications and the need for more complex databases
  - Used in specialized applications: engineering design, multimedia publishing, and manufacturing systems
- Interchanging data on the Web for e-commerce using XML
  - Extended markup language (XML) primary standard for interchanging data among various types of databases and Web pages

A Brief History of Database Applications

- Extending database capabilities for new applications
  - Extensions to better support specialized requirements for applications
  - Storage and retrieval of images, video
    - ✓ Images: scanned news or personal photographs, satellite photographic images, medical images.
    - ✓ Video: movie, news, personal digital cameras
  - Data mining applications

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